Multi-Agent System with Generative AI and Deep Reinforcement Learning for Adaptive Financial Education

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Keywords: Financial Education, Multi-Agent Systems, Deep Reinforcement Learning, Generative AI, Large Language Models, Conversational Agents.

Abstract: This paper presents a multi-agent conversational system for adaptive financial education that integrates Generative Artificial Intelligence and Deep Reinforcement Learning (DRL). The architecture combines large language models (LLMs) for intent detection and sentiment analysis with a DRL agent designed to optimize educational mission sequences based on each user's profile, engagement, and context via WhatsApp. The main contribution is the design and formalization of this integrated architecture—encompassing multi-agent orchestration, MDP-based adaptive sequencing, and deployment on a widely used messaging platform—supported by a modular, incremental implementation strategy for transcultural adaptation across Lusophone countries. We present the complete system design and report preliminary empirical results from the first implementation phase (deterministic conversational flow) with Brazilian adults, which demonstrate the system's viability and effectiveness, yielding statistically significant improvements in financial knowledge, engagement, and user satisfaction compared to traditional educational materials. These findings establish a validated baseline for the subsequent phases incorporating adaptive personalization via DRL.

1 INTRODUCTION

Financial vulnerability is a systemic challenge in Lusophone countries, where cultural, economic, and educational factors create barriers to effective financial inclusion (Faria et al., 2025). In Brazil, over 77% of families are in debt, reflecting a structural pattern of rising household financial fragility (CNC, 2025; Martins et al., 2025); Portugal exhibits household indebtedness at 82% of disposable income (Banco de Portugal, 2025); Angola faces constrained financial inclusion, with limited household access to credit and savings compared to regional peers (World Bank, 2025). These contexts—linked by the Portuguese language but differentiated by distinct socioeconomic structures—demand educational interventions that transcend mere content translation to achieve genuine cultural adaptation (Martins, 2015).

These disparities are further compounded by cultural factors that shape financial decision-making differently across Lusophone nations. In Brazil, informal economic practices and regional income inequality create unique challenges for financial literacy programs (Vieira et al., 2025). In Angola, the prevalence of informal economic sectors requires approaches grounded in local financial realities (Lopes, 2014). Portugal's integration into the European financial system creates yet another set of challenges. Traditional financial education programs—typically based on static printed materials or classroom instruction—have shown limited effectiveness in reaching vulnerable populations, particularly those with lower educational attainment (García Llamosas and Pérez-Oleaga, 2025).

Recent advances in Generative AI, particularly LLMs, have opened new possibilities for scalable, personalized educational systems (Gruenhagen et al., 2024). Conversational AI enables real-time, interactive learning experiences that can adapt to individual needs and contexts. However, most existing implementations rely on tightly coupled architectures with limited adaptability and no mechanisms for adaptive mission sequencing based on individual learner trajectories.

This paper proposes a multi-agent conversational system that combines: (i) Generative AI (Llama 3.1 (Grattafiori et al., 2024), orchestrated via LangGraph (Pattnayak and Bohra, 2025)) for natural financial education dialogues via WhatsApp; (ii) a Deep Reinforcement Learning (DRL) framework, formalized as a Markov Decision Process (MDP), designed to optimize personalized educational mission sequences based on multi-dimensional user state and a composite reward function; and (iii) a modular architecture designed for transcultural adaptation to Lusophone countries. Implementation follows an incremental strategy: the current evaluation covers Phase 1 (deterministic conversational flow), which establishes the baseline infrastructure and validates the multi-agent pipeline, while DRL-based adaptive sequencing is designed for subsequent phases.

The contribution is threefold: (1) the design and formalization of a novel multi-agent architecture integrating LLMs with DRL for adaptive financial education, including an MDP formulation with a multi-dimensional reward function capturing completion, engagement, learning, and cultural alignment; (2) an incremental implementation strategy that decouples the multi-agent conversational pipeline from the DRL optimization, enabling independent validation of each layer; and (3) preliminary empirical evidence from the Phase 1 deployment in a controlled study with 57 Brazilian adults, demonstrating that the multi-agent conversational approach yields statistically significant improvements over traditional materials across all measured dimensions, thereby validating the foundational layer upon which DRL-based personalization will operate. This work aligns with UN Sustainable Development Goals (United Nations, n.d.), particularly SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities), contributing to poverty reduction through financial capability building (Grewal et al., 2024).

The remainder of this paper is organized as follows: Section 2 reviews related work in AI-based financial education and multi-agent systems; Section 3 presents the methodology, including the MDP formalization and multi-agent architecture; Section 4 describes the preliminary empirical evaluation; Section 5 discusses the results; and Section 6 concludes with future work directions.

2 RELATED WORK

2.1 AI and LLMs in Financial Education

The application of AI to financial education has evolved significantly. Early chatbot-based systems provided rule-based guidance with limited personalization (Ramjatton et al., 2021). Recent work leverages LLMs for more sophisticated interactions: (Prasongpongchai et al., 2024) reported increased motivation with AI-generated virtual instructors in controlled experiments, while (Gómez Cano and Zapata, 2025) demonstrated ChatGPT's ability to personalize entrepreneurship and financial literacy content according to cognitive level in virtual learning environments. (Yagi et al., 2025) explored prompt engineering for generating accessible financial content in Easy Japanese, highlighting the importance of linguistic adaptation for financial inclusion. In the domain of AI-driven financial personalization, (Tandon and Bansal, 2025) examine how generative AI can empower financial access through intelligent algorithms, and (Bhattacharya et al., 2024) integrate behavioral science with AI for motivational personalization in sustainable finance advisory. (Ozili, 2025) provides a broader perspective on how AI contributes to digital financial inclusion. The cognitive dimension is addressed by (Moşoi et al., 2025), who investigate the interplay between AI assistance and students' cognitive abilities in problem-solving tasks. However, these approaches rely on single-system designs and lack adaptive sequencing mechanisms based on individual learner trajectories.

2.2 Multi-Agent Systems and DRL in Education

Multi-agent architectures allow separation of concerns—intent detection, sentiment analysis, and response generation—into modular, separately tunable components (Xu et al., 2024). Intent detection draws on transfer learning for multi-task query understanding (Shah and Siskind, 2021), while sentiment analysis employs cross-modal attention mechanisms for financial opinion integration (Liu and Yang, 2025). For underrepresented languages, recent advances in low-resource sentiment analysis (Zanga et al., 2025) are particularly relevant for Lusophone contexts. In financial domains, multi-agent systems have been applied to data integration (Chen et al., 2025a) and information retrieval (Chen et al., 2025b).

Deep Reinforcement Learning provides a well-established framework for sequential decision-

making under uncertainty (Sutton and Barto, 2018; Bellman, 1957). In education, DRL has been applied to adaptive learning path optimization (Du et al., 2025): (Yu, 2025) employs Q-learning for personalized course recommendation, demonstrating high accuracy in matching learners to content based on interaction history; (Li, 2025) combines knowledge graphs with an actor-critic architecture for dynamic learning path optimization in blended environments; and (Wang et al., 2026) proposes IB-GRPO, aligning LLM-generated learning path recommendations with educational objectives through group relative policy optimization. At the intersection of knowledge modeling and recommendation, (Ozyurt et al., 2024) integrates knowledge tracing with semantic grounding for personalized exercise recommendation, while (Qiao, 2024) combines BERT embeddings with DQN for adaptive information retrieval in educational contexts. For adaptive feedback, (Dongre et al., 2025) employs graph attention networks with reinforcement learning for student performance prediction, achieving improved personalization over traditional approaches. RLHF techniques further provide mechanisms for aligning model behavior with human preferences (Casper et al., 2023). However, these works focus on general education; applications of DRL to financial education remain scarce. Cultural factors significantly influence financial behavior (Opoku-Okuampa, 2024; Vieira et al., 2025), yet existing systems rarely incorporate cultural adaptation mechanisms. Recent work has begun exploring AI-driven cultural and linguistic adaptation: (Nouzri et al., 2025) present a multi-agent LLM architecture for Luxembourgish language learning that dynamically adapts content to a low-resource linguistic context; (Uchoa et al., 2026) propose a multi-stakeholder alignment framework for LLM-powered tutoring systems, addressing value conflicts across culturally diverse learner populations; and (Ning et al., 2025) develop an intent-aware bilingual summarization framework that demonstrates cross-lingual generalization between English and Chinese educational texts. Despite these advances, two significant gaps remain: (1) existing DRL-based educational systems focus on general education and have not been applied to financial literacy, where cultural context strongly influences learning outcomes; and (2) no current architecture combines multi-agent orchestration with DRL-based adaptive sequencing while incorporating explicit mechanisms for transcultural adaptation. Our work addresses these gaps by integrating a multi-agent conversational pipeline with DRL-based adaptive mission sequencing, deployed on WhatsApp for accessibility, within an architecture specifically designed for transcultural expansion across Lusophone countries (Martins, 2015; Banco de Portugal, 2018; Lopes, 2014).

3 METHODOLOGY

3.1 Objectives

The main objective here is to develop a multi-agent conversational system that integrates Generative AI and DRL to provide personalized, culturally adapted financial education via WhatsApp. The specific objectives are: (1) formalize the educational problem as a Markov Decision Process (MDP) with a multi-dimensional reward function that captures completion, engagement, learning, and cultural alignment; (2) implement a multi-agent architecture that separates intent detection, sentiment analysis, and response generation into modular components that can be improved independently; (3) design a DRL agent based on Deep Q-Networks (DQN) for adaptive mission sequencing that learns optimal pedagogical strategies from interaction data; and (4) validate the approach through a controlled empirical evaluation comparing the conversational system with traditional static materials, using standardized statistical methods. To ease reading and in what follows, we provide a discursive presentation of the proposed system. The complete system architecture is presented in Figure 1.

3.2 MDP Formalization

The adaptive mission sequencing problem is formalized as a Markov Decision Process (MDP) (Bellman, 1957; Sutton and Barto, 2018), a classical framework for sequential decision-making under uncertainty. The MDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where:

- $\mathcal{S}$ (**State space**): a multi-dimensional representation of the learner at each time step t, encoding financial proficiency, engagement level, interaction history, sentiment, detected intent, cultural context (Opoku-Okuampa, 2024), and the set of already completed missions. This representation—informed by advances in semantically-grounded knowledge tracing (Ozyurt et al., 2024)—enables the agent to account for both the learner's current knowledge and affective state.
- $\mathcal{A}$ (**Action space**): the selection of the next educational mission from the set of missions not

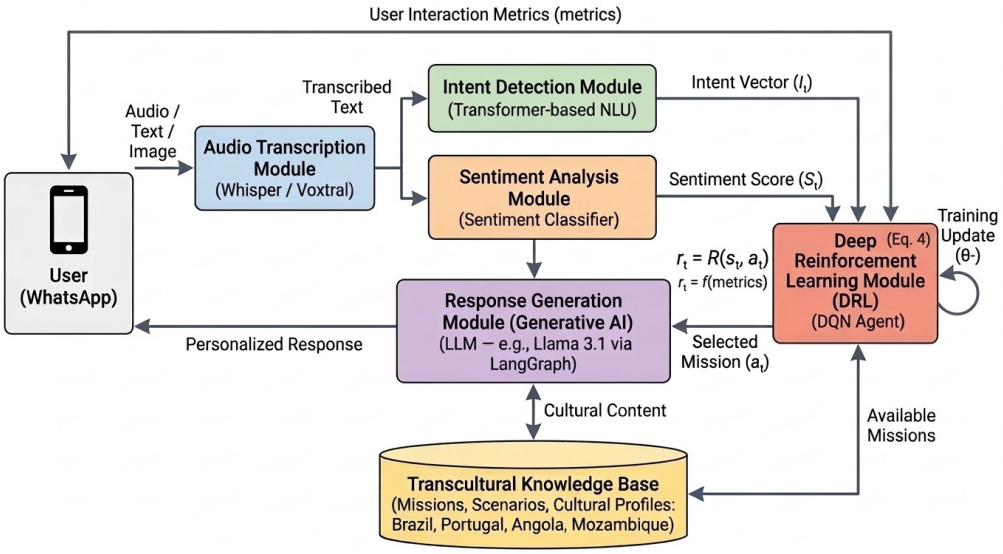


Figure 1: Architecture of the conversational agent system for financial education.

yet completed, ensuring the learner progresses through the curriculum without repetition.

- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ (**Transition function**): captures the stochastic dynamics of learner behavior—given a state and an action (mission assignment), the probability of transitioning to a new state reflects the inherent uncertainty in how learners respond to educational interventions (Du et al., 2025).
- $\mathcal{R} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ (**Reward function**): a weighted combination of four complementary signals—mission completion, engagement, learning outcomes, and cultural alignment—each capturing a distinct pedagogical objective (Dongre et al., 2025). This multi-dimensional structure ensures optimization not only for task completion but also for sustained engagement, knowledge acquisition, and cultural relevance.
- $\gamma \in [0, 1]$ (**Discount factor**): balances short-term and long-term optimization, encouraging the agent to consider the cumulative effect of mission sequences on the learner's trajectory.

To adhere to Reproducibility and Open Science Practices, the complete mathematical formalization, including all equations and parameter definitions, is available in the supplementary material[1].

[1]Full mathematical formulations: https://github.com/paulocob/article-multi-agent-financial-education

3.2.1 Optimization with Deep Q-Network (DQN)

The DRL agent employs a Deep Q-Network (DQN) (Mnih et al., 2015) to learn an optimal policy for mission selection. The core idea is to approximate the optimal action-value function using a deep neural network, which maps learner states to expected cumulative rewards for each available mission. This approach enables the agent to generalize across similar learner profiles and to scale to large state and action spaces that would be intractable with tabular methods. Training follows the temporal-difference paradigm (Sutton and Barto, 2018): the network is updated by minimizing the discrepancy between its current value estimates and bootstrapped targets computed from observed rewards and successor states. A separate target network with periodically frozen weights provides stable learning targets, mitigating the instability inherent in combining function approximation with bootstrapping. Additionally, experience replay—where past interactions are stored and sampled uniformly for training—breaks temporal correlations and improves sample efficiency. To further enhance stability and performance, the architecture can incorporate extensions such as Double DQN (Van Hasselt et al., 2016) (which reduces overestimation bias), Dueling DQN (Wang et al., 2016) (which separately estimates state value and action advantage), and Prioritized Experience Replay (Schaul et al., 2016) (which preferentially samples more informative transitions).

The mission selection policy follows an ε-greedy

strategy (Sutton and Barto, 2018) that balances exploration and exploitation: with probability $1-\varepsilon$, the agent selects the mission with the highest estimated value, while with probability ε, it explores by choosing a random mission. The exploration rate ε decays exponentially over time, allowing the agent to progressively shift from broad exploration to confident exploitation as it accumulates experience. Again, the detailed equations for the Q-value approximation, loss function, target computation, and policy are provided in the supplementary formulations.

3.2.2 Multi-Agent Generative AI System

The system employs multiple LLM-based agents, each responsible for a specific aspect of the interaction, drawing on attention mechanisms (Vaswani et al., 2017) and parameter-efficient fine-tuning via LoRA (Hu et al., 2021). The generative conversational agent produces natural-language responses conditioned on the user's input and conversation history, using an autoregressive language model (Llama 3.1 (Grattafiori et al., 2024)) built on state-of-the-art NLP infrastructure (Wolf et al., 2020) that generates each token sequentially based on all preceding tokens and the input context.

Intent detection classifies the user's communicative intent from their input using a natural language understanding (NLU) module (Shah and Siskind, 2021; Ning et al., 2025). The model produces a probability distribution over a predefined set of intent categories by projecting the contextual embedding of the input through a softmax classification layer, and the most probable intent is selected. Sentiment analysis operates in parallel, producing a probability distribution over positive, neutral, and negative sentiment classes (Liu and Yang, 2025; Zanga et al., 2025). A scalar sentiment score, computed as the difference between positive and negative probabilities, is fed into the DRL state representation, providing the reinforcement agent with real-time affective feedback about the learner's emotional state.

The integration between NLP modules and the DRL agent occurs through three mechanisms (Xu et al., 2024; Nouzri et al., 2025). First, the outputs of intent detection and sentiment analysis are combined with contextual embeddings into a unified feature vector that serves as input to the reinforcement agent. Second, the DRL state is dynamically updated at each interaction step to incorporate the latest NLP observations alongside mission outcomes and changes in the learner's proficiency. Third, the base reward signal is augmented with a contextual alignment bonus that measures how well the selected mission matches the learner's current intent and affective state, using cosine similarity between mission embeddings and the NLP feature vector. In advanced configurations, all modules—DQN, intent detection, and sentiment analysis—can be jointly optimized through a combined loss function that balances reinforcement learning and NLP objectives. The complete formalization of these components is available in the supplementary material.

3.3 Culturally Adapted Content Development

The development of financial education missions follows an iterative and collaborative process, involving experts in financial education. The current implementation targets the Brazilian context, with the methodology designed for future extension to other Lusophone cultures. The process comprises four steps: **(1) Cultural Mapping**, identifying cultural, social, and economic aspects relevant to financial education in the target context; **(2) Scenario Creation**, developing scenarios and case studies that reflect the financial realities of the target population; **(3) Mission Adaptation**, incorporating these cultural aspects into the financial education missions; and **(4) Expert Validation**, where content is reviewed by local experts in financial education and culture (Kolb, 1984; Chilikina et al., 2021).

3.4 Metrics Definition

System evaluation relies on three complementary metric families: **engagement** (DAU/MAU ratio, session duration, retention rate), **mission completion** (completion rate, quiz scores, per-stage dropout), and **satisfaction** (NPS, CSAT, qualitative feedback). For *offline* evaluation, an aggregate metric R combines these indicators as given in Equation (1):

$$R = 1.0 + \left(\frac{\text{quiz}}{100}\right) \times 1.5 + b_{\text{eng}} + b_{\text{sat}} - p_{\text{delay}} \quad (1)$$

where $b_{\text{eng}} \geq 0$ is an engagement bonus, $b_{\text{sat}} \geq 0$ a satisfaction bonus, and $p_{\text{delay}} \geq 0$ a delay penalty. This metric is distinct from the real-time reward used during DRL training: while the training reward guides the agent's policy optimization, R serves as a composite indicator for post-hoc system analysis and comparison across experimental conditions.

3.5 Implementation Approaches

Implementation follows an incremental strategy in three phases. **Phase 1 (Deterministic Flow)** uses LangGraph with pre-defined state graphs mapping

intents to responses, establishing the basic infrastructure and serving as *baseline* for data collection. **Phase 2 (Personalization)** introduces Multi-Armed Bandits (MAB) over LangGraph, where each arm represents a pedagogical strategy and the system explores alternatives to maximize engagement and learning. **Phase 3 (Complete DRL)** implements the full Deep Q-Network architecture: LangGraph manages the high-level flow while DQN selects optimal educational actions based on complete learner state, using stabilization techniques (*experience replay*, *fixed Q-targets*, Double DQN) with initial *offline* training and subsequent *online* refinement.

Algorithm 1 formalizes the Phase 1 orchestration pipeline implemented in LangGraph. A *Root Agent* receives each user message, classifies it into one of five intents, and routes it to the corresponding specialized sub-agent. In the current deterministic implementation, intent classification relies on keyword matching and few-shot prompting of the LLM. As noted in Section 4.3, misclassification of user intent was the primary source of perceived hallucinations (11% of interactions). Phase 2 and 3 will replace the deterministic router with a DRL-based policy (π_θ) that leverages the full conversational state—including dialogue history, user profile, and mission progress—to select the optimal routing action, thereby reducing intent misclassification through context-aware sequential decision-making.

Input: User message m_t, conversation state s_t
Output: Agent response r_t, updated state s_{t+1}
Root Agent receives (m_t, s_t);
$intent \leftarrow \text{CLASSIFYINTENT}(m_t, s_t)$;
// Phase 1: deterministic
// Phase 2--3: $intent \leftarrow \pi_\theta(s_t)$ via DRL policy
if $intent = \text{STARTMISSION}$ **then**
  $r_t \leftarrow \text{MISSIONAGENT}(m_t, s_t)$;
  Present next lesson module; update progress;
else if $intent = \text{DATACOLLECTION}$ **then**
  $r_t \leftarrow \text{ONBOARDINGAGENT}(m_t, s_t)$;
  Collect user profile data (income, goals, context);
else if $intent = \text{COLLECTFEEDBACK}$ **then**
  $r_t \leftarrow \text{FEEDBACKAGENT}(m_t, s_t)$;
  Record satisfaction ratings and qualitative feedback;
else if $intent = \text{ASKQUESTION}$ **then**
  $r_t \leftarrow \text{QAAGENT}(m_t, s_t)$;
  Generate contextual answer via RAG over lesson content;
else if $intent = \text{FINANCIALANALYSIS}$ **then**
  $r_t \leftarrow \text{ANALYSISAGENT}(m_t, s_t)$;
  Produce personalized financial diagnostic from user data;
end
$s_{t+1} \leftarrow \text{UPDATESTATE}(s_t, intent, r_t)$;
return r_t, s_{t+1};

Algorithm 1: LangGraph Multi-Agent Orchestration Pipeline

3.6 Evaluation and Experimentation

The conversational agent evaluation is structured in three phases:

1. **Preliminary Empirical Evaluation:** Controlled exploratory study with a restricted group of participants in the initial target context (Brazil);
2. **Controlled Case Study:** Comparison of agent performance with traditional approaches, measuring Financial Literacy, Engagement, User Satisfaction, and Cultural Adaptation.
3. **A/B Testing and Online Training:** 90% of users are exposed to the deterministic *baseline* and 10% to the new DQN agent logic; engagement, completion, and satisfaction metrics are compared; and DQN cohort exposure is progressively increased while the agent is trained *online*.

3.7 Tools and Technologies

The implementation uses a consolidated *open-source* ecosystem: Python as base language; PyTorch (Paszke et al., 2019) and TensorFlow (Abadi et al., 2016) for deep learning; Hugging Face *Transformers* (Wolf et al., 2020) for access to pre-trained models and fine-tuning. LLMs include Llama 3.1 (Grattafiori et al., 2024) and other open-source models adapted to Brazilian Portuguese. The choice of Llama 3.1 over proprietary alternatives (e.g., GPT-4) was driven by two factors: cost sustainability for deployment in low-resource contexts, and data privacy considerations given the sensitive financial information shared by users. Audio transcription employs Whisper (Graham and Roll, 2024) and Voxtral (Liu A.H. et al., 2025), enabling voice-based interaction critical for users with limited literacy. Communication occurs via WhatsApp Business API (Meta Platforms, n.d.), chosen for its near-universal adoption among the target population in Brazil (99% smartphone penetration in the 18–45 age group). Data persistence uses PostgreSQL (PostgreSQL, n.d.) for structured storage of user profiles, interaction histories, and learning analytics.

4 PRELIMINARY EMPIRICAL EVALUATION

4.1 Study Design and Participant Profile

A between-subjects evaluation of the Phase 1 system (LangGraph + Llama 3.1, via WhatsApp) compared an **experimental group** ($n = 45$) using the conversational agent with a **control group** ($n = 12$) receiving SEBRAE booklets on financial planning (SEBRAE, 2013; SEBRAE, n.d.a). SEBRAE (*Serviço Brasileiro de Apoio às Micro e Pequenas Empresas*) is Brazil's primary institution for micro and small enterprise development, founded in 1972 and currently reaching over 10 million businesses annually (SEBRAE, n.d.b). Its educational materials on financial planning are widely adopted as reference resources by public agencies, NGOs, and educational programs nationwide, making them the ecologically valid baseline—i.e., the standard intervention that the target population would typically receive in the absence of the proposed system. While this comparison inherently confounds the delivery medium (interactive vs. static) with the pedagogical approach, this design reflects the real-world alternative and thus directly addresses the practical research question of whether conversational AI can outperform the current standard practice in financial education for vulnerable populations. Participants were adults from Northeast Brazil (18–45 years, elementary/high school education) who provided informed consent prior to participation, including authorization for anonymized data collection and analysis. From the recruited pool of 57 volunteers, **45 were randomly selected** for the experimental group; the remaining 12 participants were allocated to the control group. The assigned mission was mapping personal income/expenses and generating a cash flow report. All participants completed a 10-item pre/post financial knowledge questionnaire covering budgeting, cash flow analysis, and basic financial planning concepts. The questionnaire items were designed to assess both declarative knowledge (e.g., definition of cash flow) and procedural knowledge (e.g., calculating monthly savings capacity). Content validity was established through review by two financial education specialists, and internal consistency was adequate (Cronbach's $\alpha = 0.78$). Table 1 summarizes the participant profile.

Table 1: Participant profile – Preliminary empirical evaluation.

Characteristic	Experimental	Control
Participants	45	12
Age range	18–45 yrs	18–45 yrs
Region	NE Brazil	NE Brazil
Education	Elem./High	Elem./High
Gender (M/F)	60/40%	60/40%
Medium	WhatsApp	SEBRAE booklets
Avg. duration	35 min	self-paced

4.2 Results

Of the 45 experimental participants, 90% (40) reported feeling motivated to engage with the financial education content, all of whom completed the assigned mission (**100% completion among engaged users**; 88.9% overall). This high completion rate among motivated participants suggests that once initial engagement is achieved, the conversational format effectively sustains learner attention through the full mission duration.

Sessions averaged 35 minutes and 37 messages (18 from the agent, 19 from the user), indicating balanced bidirectional interaction rather than passive content consumption. The mean satisfaction score was **4.7/5.0** (94%), reflecting strong user acceptance of the conversational approach. Notably, 78% of participants confirmed full resolution of their financial doubts during the interaction, demonstrating the system's capacity for real-time clarification of concepts that static materials cannot address.

The 11% perceived hallucination rate reported by non-motivated participants was traced to **intent misclassification** rather than factual errors in the LLM's generated content. In all identified cases, the intent detection module incorrectly routed user queries—e.g., classifying a financial doubt as a feedback or registration intent—causing the agent to respond with contextually inappropriate content. This finding highlights that in Phase 1's deterministic flow, the rigid intent-to-response mapping amplifies classification errors. The DRL-based adaptive sequencing planned for Phases 2–3 directly addresses this limitation by incorporating conversational state and user context into action selection, enabling the agent to recover from misclassified intents dynamically. Figure 2 summarizes the key outcome indicators for the experimental group.

Learning was quantified using Hake's normalized gain (Hake, 1998): $\langle g \rangle = (\text{post} - \text{pre})/(100 - \text{pre})$. The between-group comparison, including learning gains and all other outcome indicators, is presented in Section 4.3.

4.3 Between-Group Comparison

Table 2 and Figure 3 summarize the between-group comparison across all measured dimensions. The

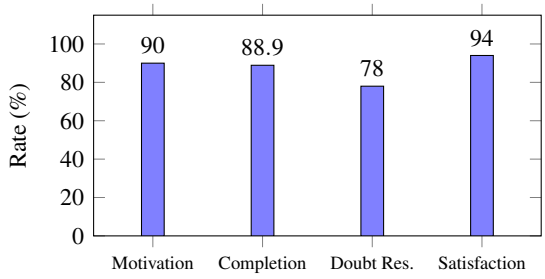


Figure 2: Experimental group ($n = 45$) outcome indicators.

conversational agent consistently outperformed the static SEBRAE booklets used by the control group, with advantages ranging from +17.0 pp (learning gain) to +53.0 pp (doubt resolution).

Table 2: Outcome comparison: experimental vs. control.

Metric	**Experimental**	**Control**
Motivation	90.0% (40/45)	58.3% (7/12)
Completion	88.9% (40/45)	50.0% (6/12)
Doubt resolution	78.0% (34/44)	25.0% (3/12)
Pre-test (%)	42.0	40.8
Post-test (%)	71.5	53.3
Learning gain (pp)	+29.5	+12.5
Norm. gain $\langle g \rangle$	0.51 (medium)	0.21 (low)
Satisfaction	4.7/5.0 (94%)	3.2/5.0 (64%)

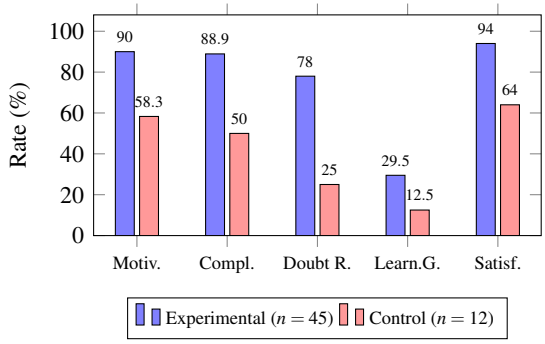


Figure 3: Between-group comparison of outcome indicators.

Advantages were observed across all metrics: motivation (+31.7 pp), completion (+38.9 pp), doubt resolution (+53.0 pp), learning gain (+17.0 pp), and satisfaction (+30.0 pp). The largest gap occurred in doubt resolution, where the conversational agent's ability to provide real-time interactive clarification was inherently unavailable in the static SEBRAE booklets used by the control group.

All between-group differences reached statistical significance ($\alpha = 0.05$), and remained significant after Holm–Bonferroni correction for five simultaneous comparisons. For categorical variables, Fisher's exact test was applied: motivation ($p = 0.017$) and completion ($p = 0.004$). For doubt resolution, the difference was highly significant ($p < 0.001$). For continuous variables, the Mann–Whitney U test was used: learning gain ($U = 102$, $p < 0.001$, Cohen's $d = 1.46$, 95% CI of the mean difference: $[9.6, 24.4]$ pp) and satisfaction ($U = 48$, $p < 0.001$, $d = 2.52$, 95% CI: $[1.12, 1.88]$ points). Effect sizes are *large* per Cohen's conventions (Cohen, 1988), with the satisfaction effect size ($d = 2.52$) being particularly high, suggesting that the interactive conversational format generates considerably greater user satisfaction than static materials.

Although the control sample is small ($n = 12$), a post-hoc power analysis confirms that the study had adequate statistical power to detect the observed effects. With $n_1 = 45$, $n_2 = 12$, and $\alpha = 0.05$, the achieved power exceeded 0.99 for both learning gain ($d = 1.46$) and satisfaction ($d = 2.52$), computed via the noncentrality parameter $\delta = d\sqrt{n_1 n_2/(n_1 + n_2)}$. Even for a conventional large effect ($d = 0.80$), the design yields power > 0.90, above the standard 0.80 threshold (Cohen, 1988). The consistency of significant differences across all measured dimensions—with uniformly large effect sizes—provides converging evidence for the effectiveness of the conversational approach. Replication with larger, balanced cohorts is planned for Phase 2 to strengthen generalizability.

5 DISCUSSION

5.1 Comparison with Related Work

Our results extend prior findings in AI-based financial education. While (Prasongpongchai et al., 2024) reported increased motivation with AI-generated virtual instructors in a controlled setting, their system did not include adaptive mission sequencing or quantitative learning assessment. Similarly, (Gómez Cano and Zapata, 2025) demonstrated ChatGPT's ability to personalize entrepreneurship content by cognitive level, but within a tightly coupled architecture without multi-agent decomposition. As shown in Table 2, our system achieved consistently higher outcomes across all dimensions, extending these findings to a lower-education population via WhatsApp with a modular multi-agent pipeline that supports independent optimization of each component—a capability unavailable in single-system deployments.

Compared to (Ramjatton et al., 2021), who validated chatbot-based financial support but with limited personalization and no between-group comparison, our architecture introduces DRL-based adaptive se-

quencing (Phases 2–3) and provides direct between-group evidence that conversational AI statistically significantly outperforms static SEBRAE booklets across all measured dimensions (Table 2, Figure 3).

Regarding doubt resolution, (Li et al., 2025) reported higher accuracy using GPT-4o for financial chart interpretation—a fundamentally different task involving visual reasoning over structured data, rather than open-ended conversational guidance for financially vulnerable populations. Our deliberate choice of Llama 3.1 as an open-source model also provides key advantages for the target deployment context: zero API costs enabling sustainable operation in under-resourced settings, full data privacy with no external data transmission, and unrestricted fine-tuning capability. Domain-specific fine-tuning planned for Phase 2 will further enhance response quality while preserving these structural advantages that proprietary APIs cannot offer.

The perceived hallucination rate, traced to intent misclassification errors in the deterministic pipeline (Section 4), is consistent with known limitations of rigid intent-to-response mappings in conversational systems (Liu S. et al., 2025). Unlike factual hallucinations intrinsic to LLM generation, these errors stem from the Phase 1 architecture's inability to use conversational context when routing user queries. The DRL agent in Phases 2–3 will incorporate the full conversational state vector—including intent history, sentiment trajectory, and mission progress—into its policy, enabling context-aware intent disambiguation and dynamic recovery from misclassifications. Additionally, domain-specific fine-tuning and retrieval-augmented generation (RAG) will further reduce the risk of factual errors in generated content.

5.2 Contributions and Implications

This work contributes to **Conversational AI** by introducing a multi-agent architecture that separates intent detection, sentiment analysis, and response generation into separately tunable agents—combined with DRL for adaptive pedagogical personalization. Unlike single-system chatbot implementations, this decomposition allows targeted improvements to individual components without disrupting the overall system. In **Financial Education**, it provides a replicable, empirically validated model for culturally adapted delivery, with a demonstrated medium normalized learning gain (Table 2) providing quantitative evidence of knowledge acquisition among financially vulnerable populations. The exclusive use of **open-source** models (Llama 3.1, LangGraph) lowers adoption barriers in under-resourced settings, avoiding dependency on proprietary APIs that may be cost-prohibitive for organizations serving low-income populations.

From a practical standpoint, the results demonstrate that **messaging platforms already adopted by target populations** (WhatsApp) are viable channels for structured financial education, eliminating the need for dedicated apps. The large advantage in doubt resolution over static booklets (Table 2) highlights the critical role of **real-time interactive clarification** for populations with limited financial knowledge. The modular architecture allows NGOs, public agencies, and microfinance organizations to **deploy and customize** the system at low cost—particularly relevant for programs targeting the 77.6% of Brazilian families currently in debt (CNC, 2025).

The alignment with the **Sustainable Development Goals** (United Nations, n.d.; Grewal et al., 2024) strengthens the societal relevance of this contribution. SDG 4 (Quality Education) is directly addressed through the adaptive pedagogical approach, while SDG 10 (Reduced Inequalities) is targeted by deploying financial education on platforms accessible to vulnerable populations. The open-source nature of the system further supports SDG 17 (Partnerships for the Goals) by enabling replication and adaptation by organizations across different Lusophone contexts.

5.3 Limitations

As a preliminary study, this work presents limitations that are being addressed in subsequent phases. The **unbalanced sample** ($n = 45$ experimental vs. $n = 12$ control), driven by logistical constraints in recruitment, limits generalizability; however, post-hoc power analysis confirms adequate power (> 0.99) for the observed effect sizes (Section 4.3). Phase 2 will include larger and balanced cohorts.

A central limitation is that the evaluation covers only **Phase 1 (deterministic flow)**, meaning the DRL-based adaptive sequencing—a core architectural contribution—remains empirically unvalidated. The current results therefore demonstrate the viability of the multi-agent conversational pipeline, but cannot yet confirm whether DRL-driven personalization will yield additional gains over the deterministic baseline. This component will be empirically validated in Phases 2–3 through A/B testing comparing deterministic and DRL-adaptive sequencing under controlled conditions.

Furthermore, the experimental design introduces an inherent **medium confound**: the experimental group interacted with a conversational AI agent via WhatsApp, while the control group received static SEBRAE booklets. This means the observed differ-

ences may partly reflect the advantages of interactive digital delivery (real-time feedback, conversational engagement) rather than the specific multi-agent architecture. Disentangling the contribution of the delivery medium from the pedagogical approach would require additional conditions (e.g., a non-AI interactive chatbot or a static digital version of the same content), which are planned for Phase 2.

The study was conducted in a **single geographic and cultural context** (Northeast Brazil), with a specific demographic profile (18–45 years, elementary/high school education) and a single financial task (cash flow mapping). Generalization to other financial topics, educational levels, age groups, or Lusophone countries cannot be assumed from these results. Transcultural field studies in Portugal, Angola, and Mozambique are planned to assess transferability, and future evaluations will diversify the financial tasks and participant profiles. The **perceived hallucination rate**, traced to intent misclassification in the deterministic pipeline, will be mitigated through DRL-based context-aware intent disambiguation in Phases 2–3, complemented by domain-specific fine-tuning and retrieval-augmented generation (RAG). Although participants were **randomly selected** for the experimental group, control group allocation was not independently randomized, and the **unequal group sizes** may introduce residual selection bias—a limitation that will be addressed through improved experimental design. Finally, **long-term retention** of financial knowledge and behavioral change were not assessed; longitudinal follow-up studies are planned to evaluate sustained impact beyond immediate learning gains.

Regarding **threats to validity**: (i) *Internal validity* may be affected by testing effects, as the same 10-item questionnaire was administered pre and post, potentially inflating gains through item familiarity rather than genuine learning; maturation effects are unlikely given the short session duration ($\sim$35 min), but cannot be fully excluded. The Hawthorne effect—whereby participants alter behavior due to awareness of being observed—may have contributed to higher engagement in the experimental group, as the interactive WhatsApp format made the research context more salient than reading a booklet. (ii) *External validity* is constrained by the single financial task (cash flow mapping), single demographic profile (18–45, elementary/high school, Northeast Brazil), and single delivery context (WhatsApp); generalization to other financial topics, educational levels, age groups, or Lusophone countries requires further validation. (iii) *Construct validity* of the satisfaction measure may be affected by social desirability bias, as self-reported Likert scores tend to overestimate positive attitudes, particularly in face-to-face-like conversational interactions; future phases will incorporate behavioral proxies (e.g., voluntary return rate, unsolicited engagement) to complement self-report measures.

6 CONCLUSIONS AND FUTURE WORK

This paper presented a multi-agent conversational system integrating Generative AI and Deep Reinforcement Learning for adaptive financial education, formalized as an MDP with a multi-dimensional reward function. The system architecture separates intent detection, sentiment analysis, and response generation into independently optimizable agents, orchestrated via LangGraph and powered by open-source LLMs (Llama 3.1) deployed on WhatsApp. The preliminary evaluation with 57 Brazilian adults demonstrated that the system statistically significantly outperforms traditional materials across all measured dimensions (Table 2), with large effect sizes. These results validate the viability of the approach and establish a *baseline* for subsequent phases, while identifying intent misclassification—the root cause of perceived hallucinations—as the primary technical challenge to be addressed by DRL-based context-aware sequencing in Phases 2–3.

As future work, we plan: (i) transcultural expansion through field studies in Portugal, Angola, and Mozambique, leveraging the shared Portuguese language while adapting to distinct cultural and economic contexts; (ii) *transfer learning* techniques to accelerate cultural and linguistic adaptation across Lusophone countries; (iii) exploration of advanced DRL architectures (A3C, PPO) for policy optimization in the adaptive mission sequencing component; (iv) Explainable AI (XAI) mechanisms for transparency in agent recommendations, increasing user trust; and (v) larger-scale evaluation with increased and balanced sample sizes to strengthen statistical power and generalizability.

The promising results from this preliminary evaluation suggest that conversational AI systems, when properly designed with multi-agent architectures and deployed on accessible platforms, can serve as effective tools for financial education among vulnerable populations. The open-source nature of the system and its modular architecture provide a foundation for broader adoption and adaptation by educational institutions, NGOs, and public agencies across the Lusophone world and beyond.

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